

ME5412 Robotics for Healthcare

Stroke and Stroke Rehabilitation

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OUTLINE

- Definition & Epidemiology of stroke
- Consequences of stroke
- Stroke recovery Mechanism
- Clinical practices of stroke rehabilitation and assessment
- Stroke Rehabilitation Interventions
- Robotics in stroke rehabilitation



DEFINITION & EPIDEMIOLOGY OF STROKE

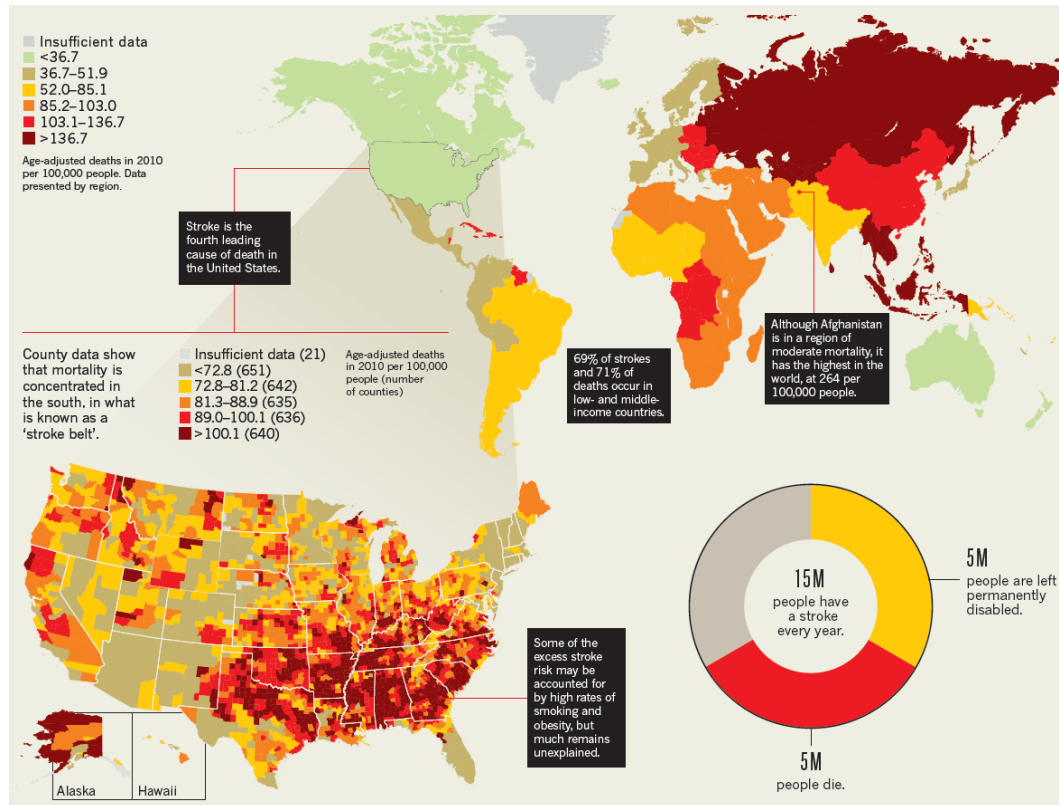
STROKE (ANIMATION)



SYMPTOMS OF STROKE

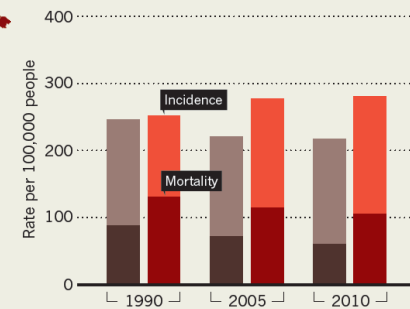


STROKE STATISTICS



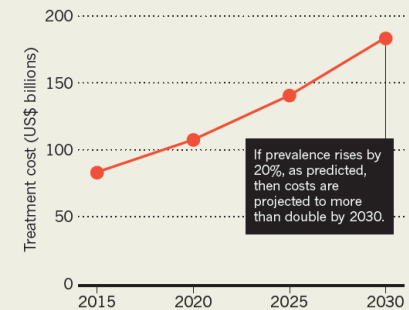
AGE CONCERN

Both incidence and mortality are falling in high-income countries. But prevalence is rising, particularly in high-income nations.

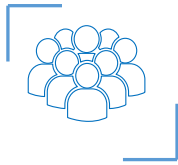


FINANCIAL TIMES

Direct medical costs are expected to rise substantially in the United States because prevalence will increase as the population ages.



STROKE STATISTICS



70 million

Worldwide stroke survivors by 2030¹



16 million

New cases worldwide per year²



30–66%

Patients exhibit motor deficits six months after stroke²



73.7 billion

Estimated direct and indirect cost of stroke in 2010 in the US³

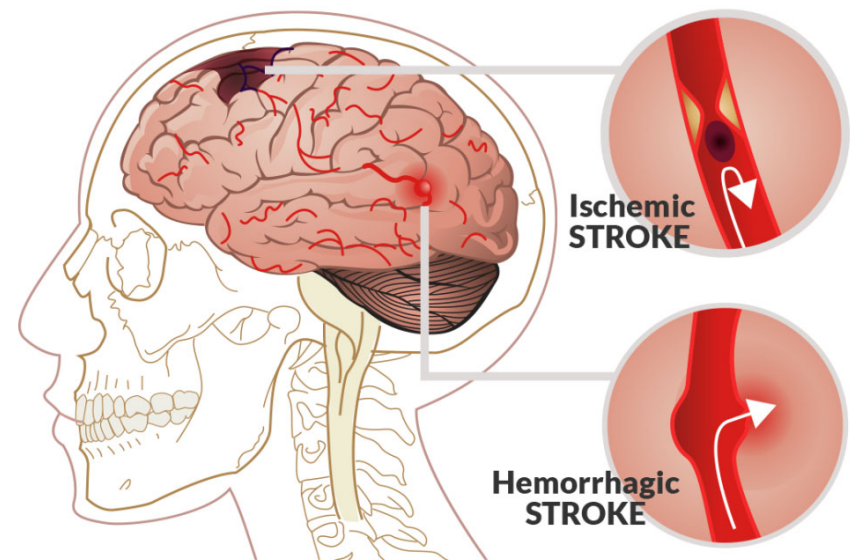
1. Feigin VL, Forouzanfar MH, Krishnamurthi R, et al. Global and regional burden of stroke during 1990–2010: findings from the Global Burden of Disease Study 2010. Lancet. 2014;383(9913):245–254.

2. Suhyun, Lee, Yumi, et al. Effect of Virtual Reality-based Bilateral Upper Extremity Training on Upper Extremity Function after Stroke: A Randomized Controlled Clinical Trial[J]. Occupational Therapy International, 2016, 23(4):357–368.

3. Giovanni M, Stefano P, Andrea C, et al. Robot-assisted gait training for stroke patients: current state of the art and perspectives of robotics[J]. Neuropsychiatric Disease & Treatment, 2017, 13:1303–1311.

TYPES OF STROKE

- **Stroke**
 - Any ischemia or hemorrhage within the central nervous system, including brain, spinal cord, or retina regardless the presence of clinically detectable symptoms
- **Types of stroke**
 - Ischemic: caused by reduced blood supply
 - Hemorrhagic: caused by disrupted blood supply
 - Transient ischemic attack (TIA):
 - a transient episode of neurological dysfunction caused by focal brain, spinal cord, or retinal ischemia, without acute infarction





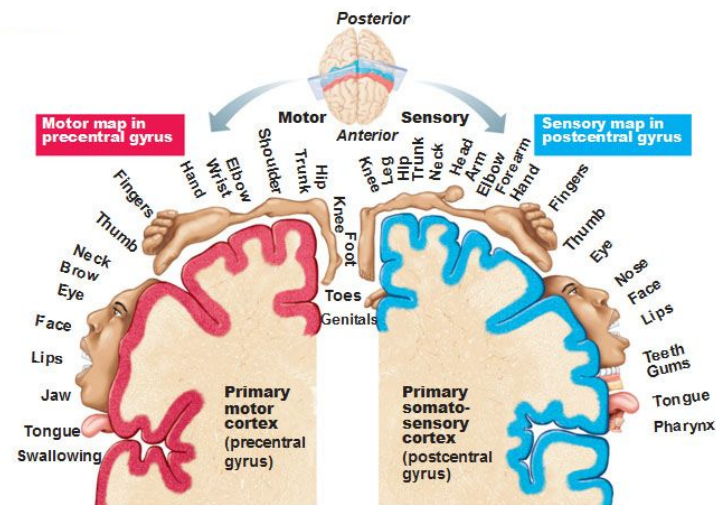
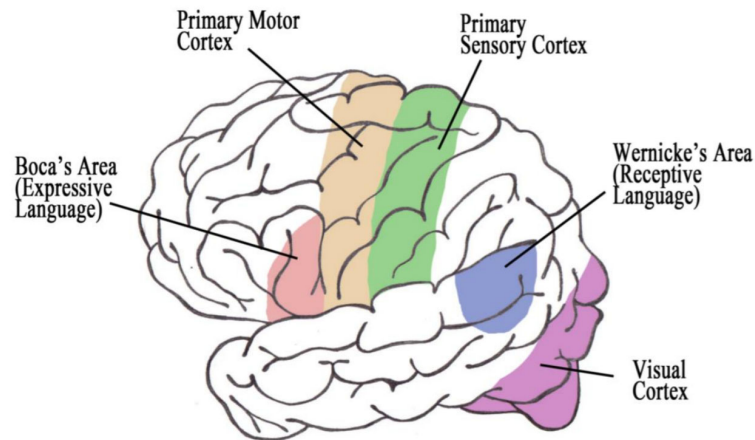
CONSEQUENCES OF STROKE

EFFECTS OF STROKE



CONSEQUENCES OF STROKE

- Lesion location determines the consequences



CONSEQUENCES OF STROKE

- **Motor function**
 - Inability to move one or more limbs on one side of the body
- **Cognitive function**
 - Inability to understand
- **Speech function**
 - Inability to formulate speech
- **Visual function**
 - Inability to see one side of the visual field (neglect)

MOTOR FUNCTION IMPAIRMENTS

- **Hemiparesis:**
 - muscular weakness or partial paralysis restricted to one side of the body (upper and lower limbs)
- **Spasticity:**
 - a condition in which a muscle involuntarily contracts
 - Spastic co-contraction: inappropriate antagonist recruitment triggered by volitional command
- **Abnormal motor synergies:**
 - loss of independent control of selected muscle groups (e.g., voluntary forward reaching with shoulder abduction and elbow flexion)



STROKE RECOVERY MECHANISM

STROKE RECOVERY

Statistics:

- 10% spontaneous recovery in 8 – 12 weeks
- 10% severe debilitating, no significant recovery
- 80% with deficits, benefit from intense rehabilitation program

STROKE RECOVERY MECHANISMS

- **Spontaneous – no rehab**
 - Collateral sprouting of axons and dendrites to make new connections
- **Neural Plasticity**
 - Reconnection of neural pathways and synapses

Once stroke occurs rehabilitation offers best opportunity for improving functional outcomes

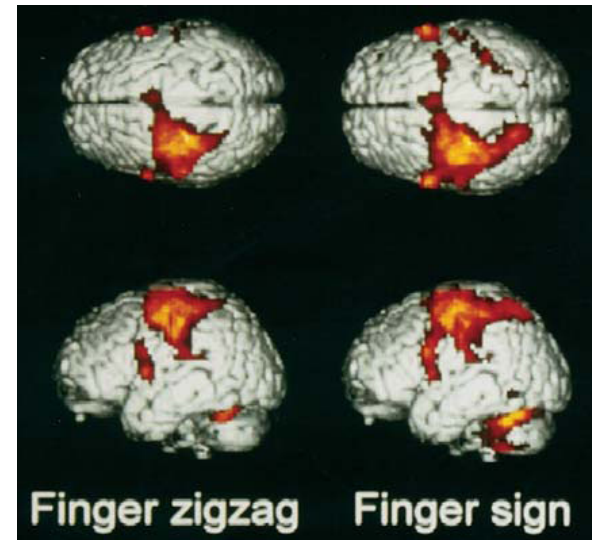
- **Why Stroke rehab works?**
 - Rehab facilitates Neuroplasticity
 - Animal and clinical evidence proved the efficacy

NEUROPLASTICITY

- Definition in (Merriam-Webster medical dictionary)
 - The capacity for continuous alteration of the neural pathways and synapses of the living brain and nervous system in response to experience of injury
- Neuroplasticity after CVA , TBI, and SCI investigated with
 - Brain imaging techniques
 - Animal models
- Neuroplasticity is much more extensive than previously thought
 - Neuroplasticity actually occurs every day of our lives, as our nervous system changes in response to experience

NEUROPLASTICITY RESEARCH

- Brain motor areas which are “not used” can generate upper limb movements if properly stimulated
- The same motor function can be activated by multiple (different, non contiguous) brain motor areas



This situation proves the flexibility of the motor output organization and is a key issue to promote functional recovery after stroke

Motor learning and plasticity can be exploited for neurorehabilitation therapy, i. e. to promote functional recovery

NEUROPLASTICITY

- Therapy training can “turn on” undamaged areas of the brain to take over function of damaged portion
 - Brain reorganize or “rewire” itself in response to changes in afferent and efferent information
- Principles in utilizing neuroplasticity:
 - Timing of therapy
 - Task specific and consistency
 - Intensity and repetition
 - Patient motivation and participation
 - Feedback

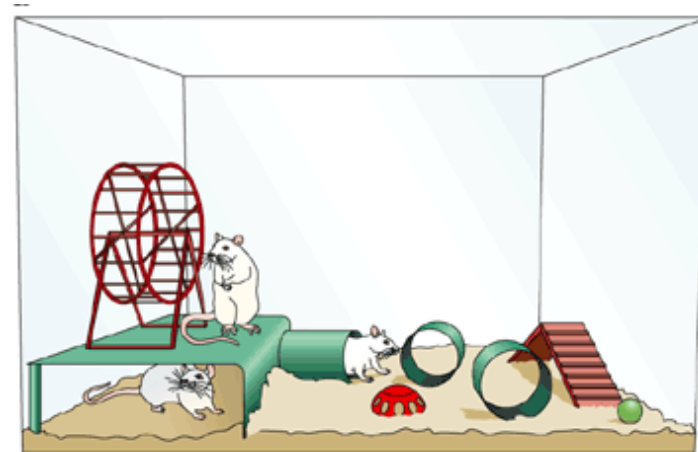
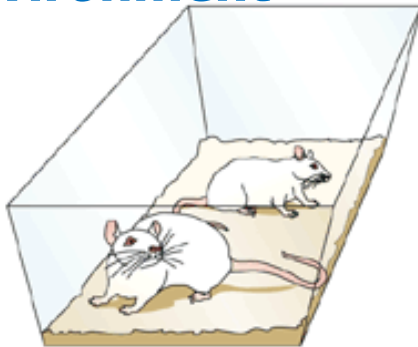
LEARNED NONUSE

- Unused body systems undergo cellular, metabolic, and structural changes which over time are interpreted by the CNS as the norm
- Learned nonuse results in decreased recovery and function of the limb
- **CONSTRAINT INDUCED MOVEMENT THERAPY (CIMT)**
 - Technique used to reverse learned nonuse
 - The unimpaired upper extremity is restrained, forcing the patient to use the affected extremity

USE IT OR LOSE IT

Rehab training (enriched environments with animals) increases cortical representation with subsequent functional recovery

In animal studies key factors promoting recovery include *increased activity and complex, stimulating environment*



Lack of rehab decreases cortical representation and delays recovery

TIMING OF THERAPY

- Brain is “primed” to “recover” early in post-stroke period
- There is a time window when brain is “primed” for maximal response to rehab therapies, such that delays are detrimental to recovery
 - Challenging patients to perform higher-order tasks as early as possible
 - Stroke patients may be able to leap-frog over lower-level activities prescribed by current traditional practice.

The Earlier the Better

ROLE OF INTENSITY OF THERAPY

- Post-stroke rehab increases motor brain reorganization, while lack of rehab reduces reorganization
- More intensive motor training further increases brain reorganization
- Clinically, greater intensity of stroke rehab therapies is associated with improved outcomes

More is Better

STROKE REHAB MUST BE TASK-SPECIFIC

- Functional reorganization of cortex greater for tasks meaningful to animal; repetitive activity not enough
- Rehab must be task-specific, focusing on tasks important and meaningful to patient



INACTIVE AND ALONE

In a conventional therapeutic day

- >50% time in bed
- 28% sitting out of bed
- 13% in therapeutic activities
- Alone for 60% of the time

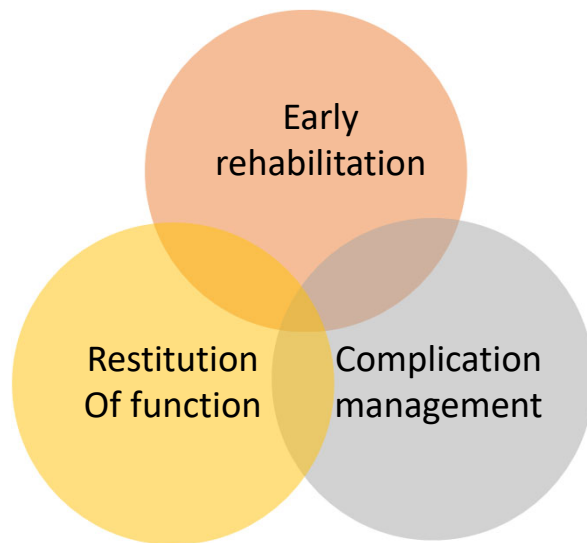
(Inactive and alone, Bernhardt et al, Stroke 2004)

Increased activity and environmental stimulation is important to neurological recovery



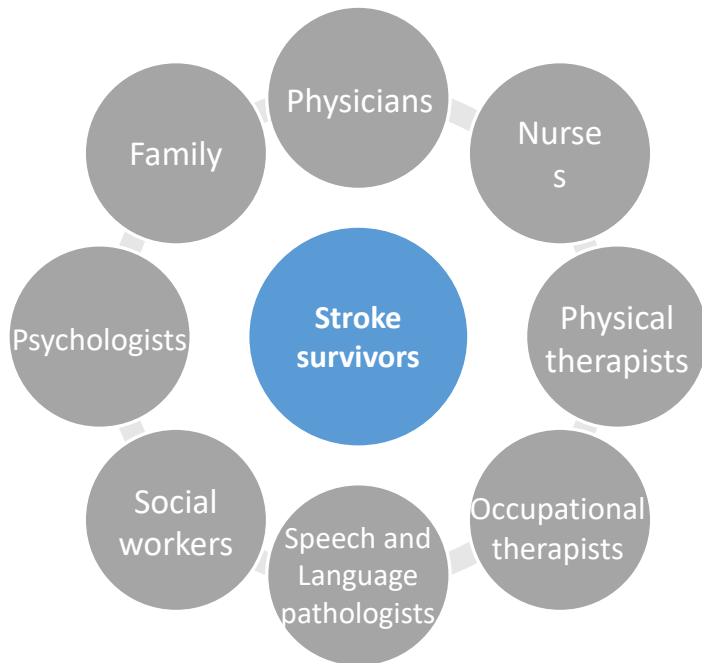
CLINICAL PRACTICES OF STROKE REHABILITATION AND ASSESSMENT

GOALS FOR POST-STROKE REHABILITATION



- Early rehabilitation
 - mobilization, mobility
- Complication management
 - vital sign control, secondary stroke/complication prevention
- Restitution of function
 - achieve the greatest possible autonomy and complete psychosocial reintegration

POST-STROKE REHABILITATION: TEAMWORK



- Physical factors
 - Cognitive effects
 - Physical effects
- Emotional factors
 - Motivation
 - Mood
- Social factors
 - Support
- Therapeutic factors
 - Early rehabilitation

OCCUPATIONAL VS. PHYSICAL THERAPY

Occupational Therapy

- A client-centered health profession concerned with promoting health and well being through occupation



Physical Therapy

- Provides services to develop, maintain and restore maximum movement and functional ability throughout the lifespan



OCCUPATIONAL THERAPY FOR STROKE



PHYSIOTHERAPY FOR STROKE



EVALUATION

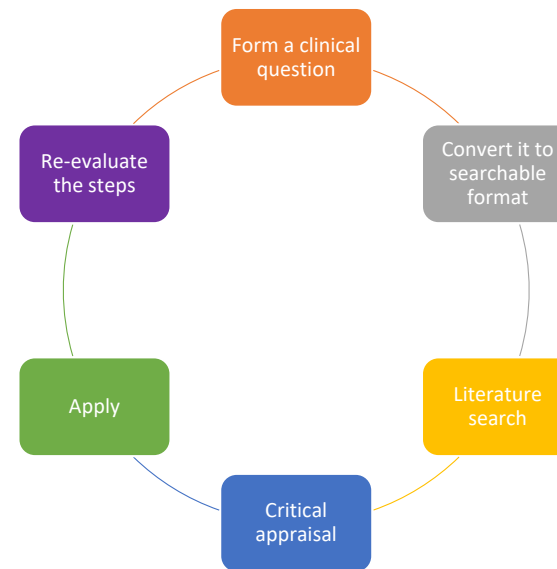
- Motor deficits
 - Spasticity
 - Muscle strength
 - Motor control
 - Coordination
 - Motor quality
- Sensory deficits
- Perceptual deficits e.g. neglect
- Cognitive deficits
- Daily function
- Language problem
- Psychological problem
- Social interaction problem

EVIDENCE-BASED PRACTICE

- The application of the principles of evidence-based medicine to clinical questions in the field of rehabilitation



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POST-STROKE MOTOR REHABILITATION

- Motor learning: a set of processes associated with practice or experience leading to relatively permanent changes in the capabilities of responding
 - Acquisition phase: immediate effects
 - Retention phase: short-/long-term effects
 - Transfer test: ability to generalize to new task

STROKE REHABILITATION OUTCOMES

- 80% Independent Mobility
- 70% Independent Personal Care
- 40% Outside Home
- 30% Work

FACTORS AFFECTING REHAB OUTCOMES

- Neurological Deficits
- Motivation Level
- Learning Ability
- Level of Emotional and Social Support
- Coping and Adaptability
- Medical Comorbidities
- Rehabilitation Training

ASSESSMENTS OF MOTOR FUNCTION

- For both upper and lower extremities
 - Fugl-Meyer Assessment (FMA)
 - Motor Assessment Scale (MAS)
 - Rivermead Motor Assessment (RMA)
 - Chedoke-McMaster Stroke Assessment (CMSA)
 - Motor Club Assessment (MCA)
 - Motricity Index (MI)
 - Stroke Rehabilitation Assessment of Movement (STREAM)



STROKE REHABILITATION INTERVENTIONS

MOTOR FUNCTION REHABILITATION

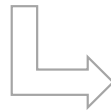
Acute/Subacute stage

- Range of motion exercise
- Positioning
- Functional mobility training
- ADL training
- Splinting
- Spasticity management



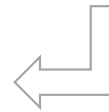
Chronic stage

- Continuum of acute/subacute phases
- Neurodevelopmental techniques
- Motor learning approaches
- Functional, task-specific overground locomotor training
- Task-oriented training

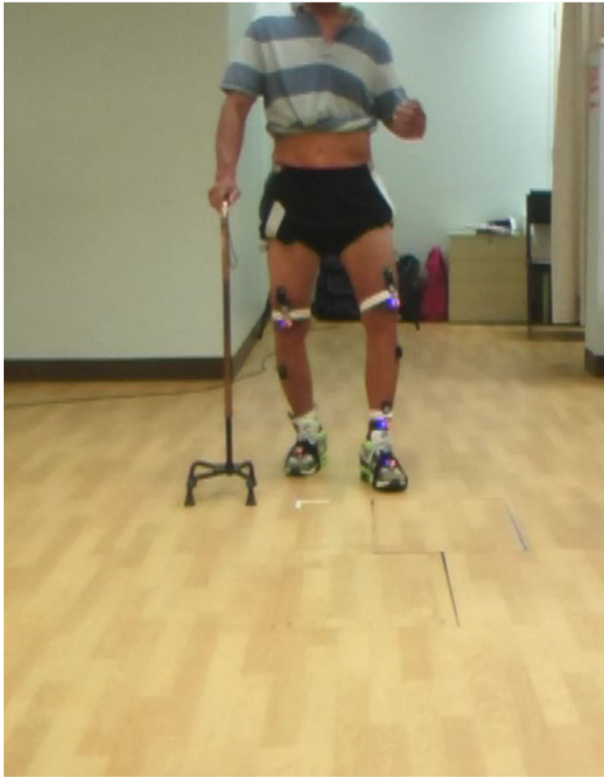


Community

- Home adaptations
- Assistive devices
- Supportive services
- Home exercise program



IMPAIRED GAIT AFTER STROKE



- Trunk compensation
- Slow and asymmetric steps
- Circumduction
- Reduced weight bearing on the affected leg
- Spastic gluteal and quadriceps muscle group
- Weak hip flexors, hamstring, and ankle dorsiflexors → extended hip and knee & drop-foot

GOALS OF GAIT REHABILITATION



- Restoration of the ability to walk
 - Trunk control
 - Balance
 - Motor control
 - Muscle performance (strength, power, and endurance)
- Independent ambulation

CONVENTIONAL GAIT REHABILITATION



Range of motion exercise



Neurodevelopmental techniques



ADL training



Balance training



Overground gait training



Spasticity management/splinting

LIMITATIONS FOR CONVENTIONAL APPROACHES



- Additional manpower
- Inconsistent assistance



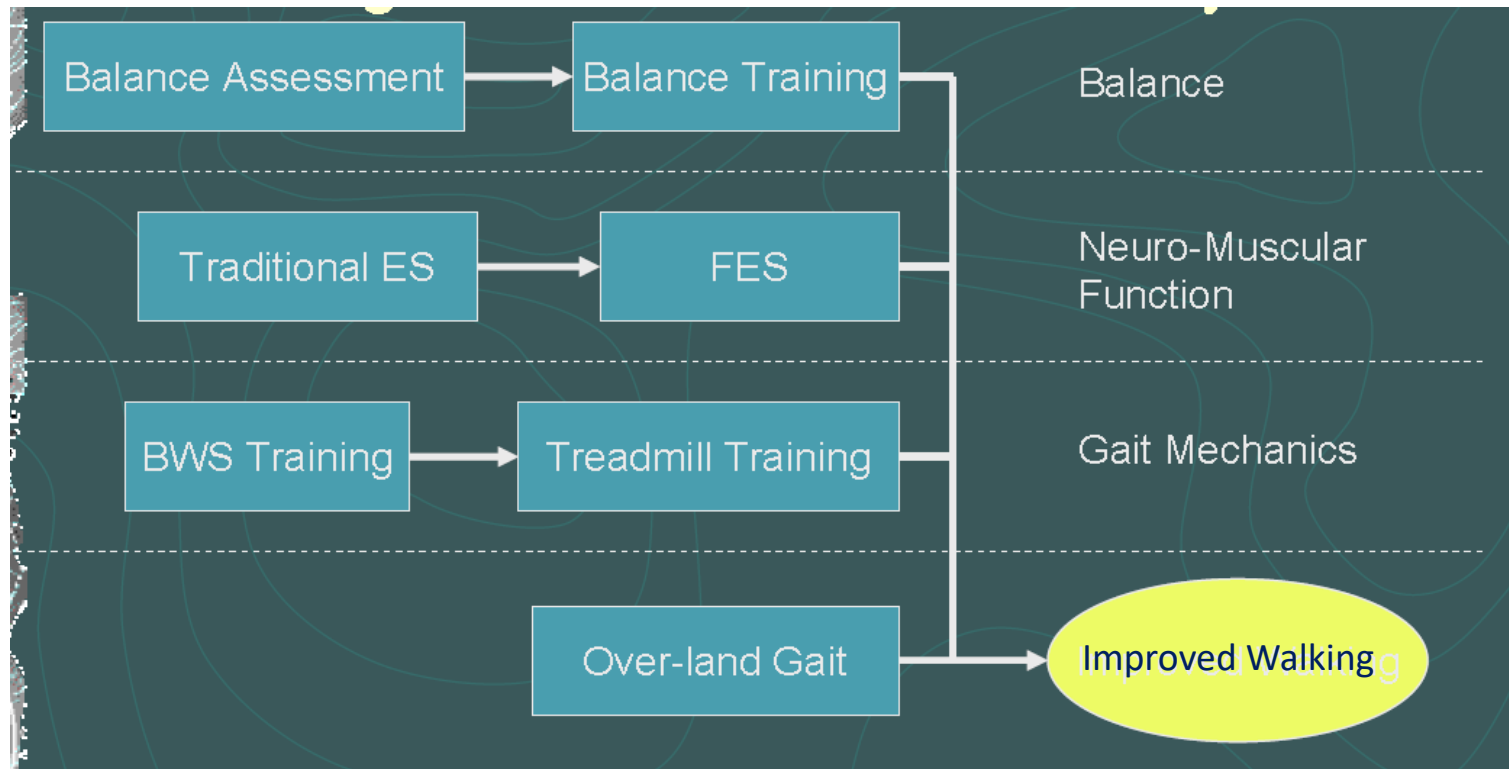
- Non-quantitative assessment

- Unable to track progression
- Less effective
 - Inefficient feedback
 - Insufficient support
 - Inconsistent assistance
- Labor intensive

LOCOMOTION REHABILITATION THERAPY

- Key components of current practice:
 - Balance assessment and training
 - Functional Electrical Stimulation (FES)
 - Body Weight Support (BWS) gait training
 - BWS treadmill gait training

LOCOMOTION REHABILITATION THERAPY



BALANCE ASSESSMENT AND TREATMENT

- Balance training allows:
 - Comprehensive evaluation of transitional and precursor movements required for normal walking.
 - Retraining of identified deficiencies.
 - Enable evidence-based training



Balance Master

FUNCTIONAL ELECTRICAL STIMULATION (FES)

- Electrical stimulation
 - Sensory electrical stimulation
 - High-frequency transcutaneous electrical nerve stimulation (TENS): electrical somatosensory stimulation of a peripheral nerve via cutaneous electrodes (80 – 100 Hz)
 - Electroacupuncture
 - Muscle electrical stimulation
 - Low-frequency transcutaneous electrical nerve stimulation: 1 – 5 Hz
 - Neuromuscular electrical stimulation (NEMS): induces muscle (10 – 50 Hz)
- Functional Electrical Stimulation (FES)
 - A technique that causes a muscle to contract through the use of an electrical current.

EVIDENCE OF FES

Upper Limbs

- Improve hand function and dexterity, motor control, motor function, functional independence, & strength

Lower Limbs

- Effective on motor control, motor function, spasticity, walking efficiency, walking speed, strength



http://www.neurolinkphysio.co.uk/images/img_arm_myotrac1.jpg

LOWER EXTREMITY FUNCTIONAL ELECTRICAL STIMULATION (FES)

- **Bioness Leg Unit:**
 - Miniature Gait sensor
 - Wireless Control Unit
 - Facilitate a more fluid gait
 - Increased walking speed



BIONESS L300 PLUS SYSTEM OVERVIEW



BODY WEIGHT SUPPORT (BWS) TRAINING

- **BWS benefits:**
 - Reducing patient effort
 - Increased training intensity
 - Improve postural retraining
 - Research validated the training efficacy



LiteGait BWS system

IMPAIRED UPPER LIMB FUNCTION AFTER STROKE



- Muscle weakness
- Sensory loss
- Pain
- Spasticity
- Abnormal motor synergies

GOALS OF UPPER LIMB REHABILITATION

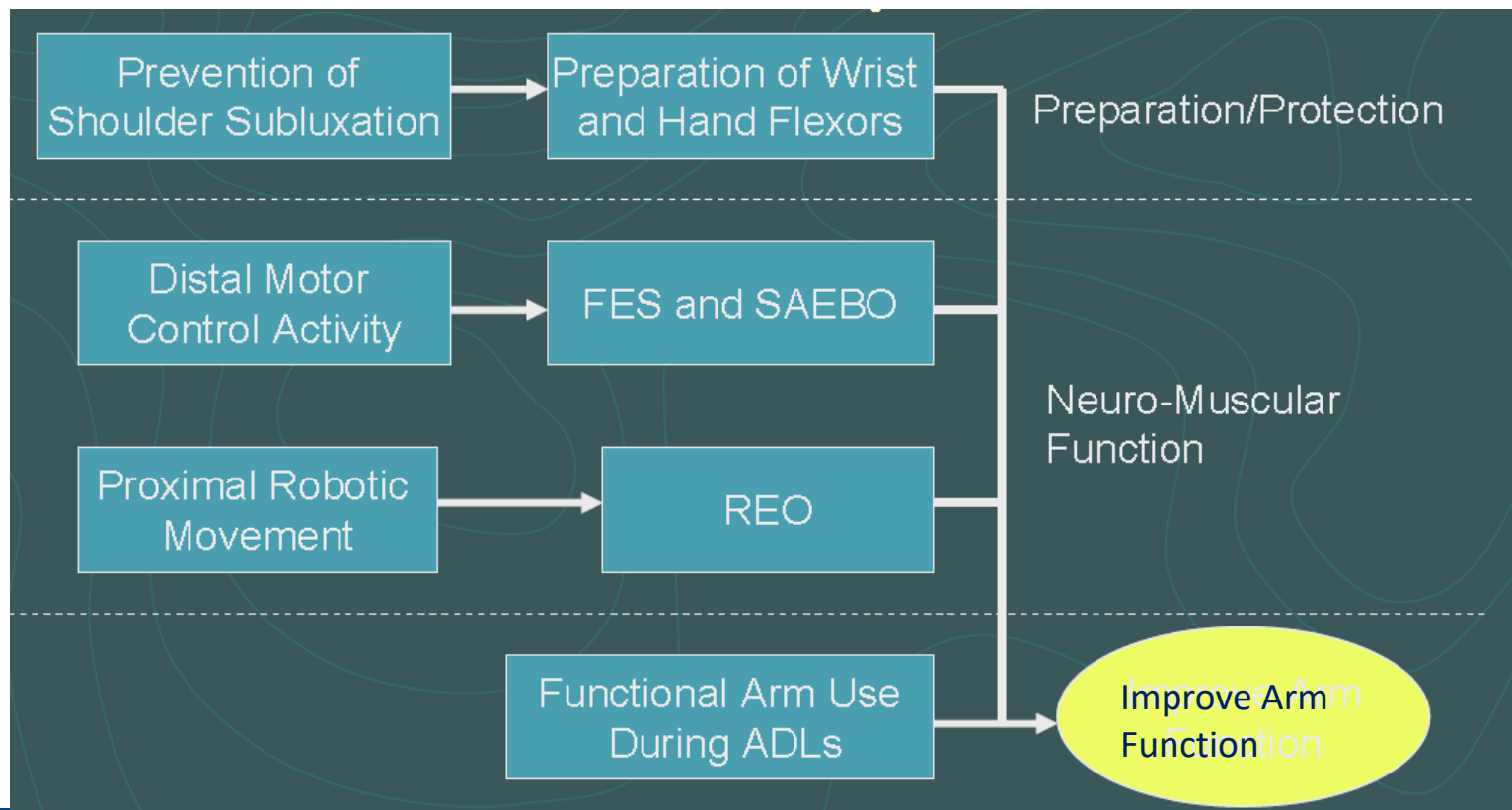


- Prevent complications
 - Immobilization
 - Muscle flaccidity
 - Shoulder subluxation
- Improve motor/sensory functions
- Return to family/community

UPPER LIMB FUNCTION RECOVERY

- Key Components
 - Joint and soft tissue protection and preparation
 - Repetitive stimulation of proximal motor control
 - Active facilitation of distal motor control
 - FES assessment and treatment

ARM FUNCTION RECOVERY



ARM FUNCTION RECOVERY -PREPARATION AND PROTECTION

- To provide:
 - Reduction of subluxation in sitting and standing
 - Reduction of shoulder pain
 - Compression of shoulder, elbow and wrist during standing and ambulation to facilitate return
- Research proved efficacy



GivMohr Slings

ARM FUNCTION RECOVERY -PREPARATION AND PROTECTION

- Addresses soft tissue shortening of the long finger flexors
 - The most limiting factor to regaining hand function following a stroke.
- Utilizes a low load long duration stretch to return the fingers to the desired position.

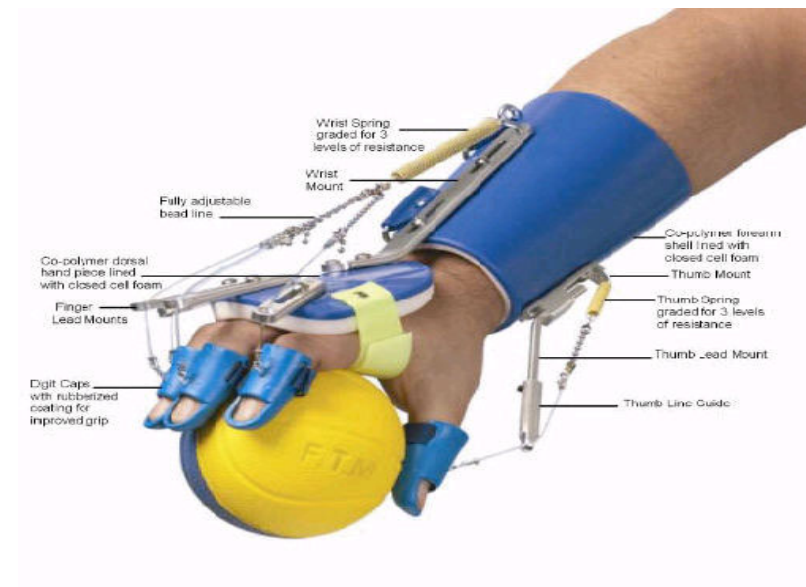


The SAEBOstretch

ARM FUNCTION RECOVERY

- Dynamic hand orthosis example:
 - Spring loaded extension system
 - Extension spring system assists in re-opening the hand to release the object
 - Assists in re-opening thumb and fingers following functional grasping
 - Non-electrically based, purely mechanical

SaeboFlex



www.saebo.com

UPPER EXTREMITY FES

- **Bioness H200 Arm Unit:**
 - The can be used both as a neuroprosthesis (i.e., as a functional aid for performing activities of daily living) and as a therapeutic device to aid motor recovery of function following a stroke.





EMERGING SOLUTIONS OF STROKE REHABILITATION

VIRTUAL REALITY (VR)

- An artificial environment which is experienced through sensory stimuli (as sights and sounds) provided by a computer and in which one's actions partially determine what happens in the environment.



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EVIDENCE OF VIRTUAL REALITY

Upper Limbs

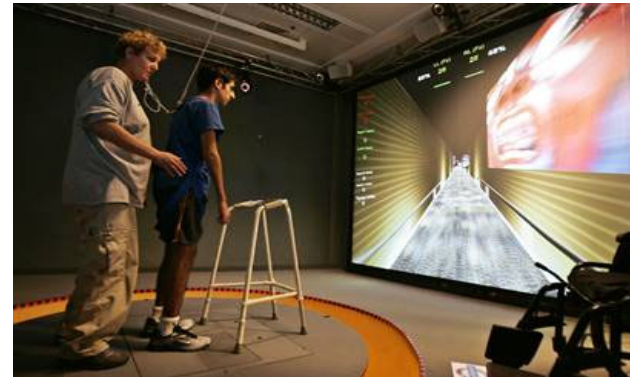
- Significant effect on arm function
- Beneficial in improving daily function
- Most effective VR program
- Patient characteristics



Virtual Reality for hand rehab

Lower Limbs

- No significant effect on gait speed
- VR + conventional therapy is more effective in improving balance and gait ability
- Participation level
- Follow-up



RHYTHMIC THERAPY- INTERACTIVE METRONOME

- Computerized program with headphones
- Computer generated reference tone
- Match the rhythmic beat with repetitive motor actions (hands or feet)
- Immediate feedback measured in milliseconds
- Score can be recorded as feedback

www.interactivemetronome.com

INTERACTIVE METRONOME

Mechanisms:

- Addresses brain timing, rhythmicity, and synchronicity
- Increases the speed and coordination of informational signals within the brain
- Improves the processing abilities that affect attention, motor planning, and sequencing
- Strengthens many fundamental learning capacities and higher level cognitive skills such as planning, organizing, and language

RHYTHMIC THERAPY-GAIT TRAINING



MUSIC-ASSISTED THERAPY



BIODEX

Biodex Music-Assisted Therapy for the Gait Trainer™ 3

ROBOT-ASSISTED THERAPY (RT)

- Using robots to facilitate recovery or compensate impairments
 - Therapeutic robots
 - End-effector-type
 - Exoskeleton-type
- Upper limbs vs. lower limbs



http://www.imalab.net/sites/default/files/image/news/MITRobot_406X248.jpg



<http://greaterkansas.mdnews.com/sites/default/files/media/vc-armeospringjanarombeck750.jpg>



<http://www.reha-stim.de/cms/assets/images/GT-I-mit-Person>



http://www.inovis.ch/en/wp-content/uploads/sites/2/2016/03/stories_Hocoma_Lokomat_bgwhite.jpg



BENEFITS OF ROBOTICS TECHNOLOGIES FOR STROKE

1. Provide high-dosage, high-intensity practice
2. Provide highly consistent practice that helps reduce variability
3. Reduce burden on occupational and physical therapists
4. Provide a mechanism for quantitative evaluation

Summary

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